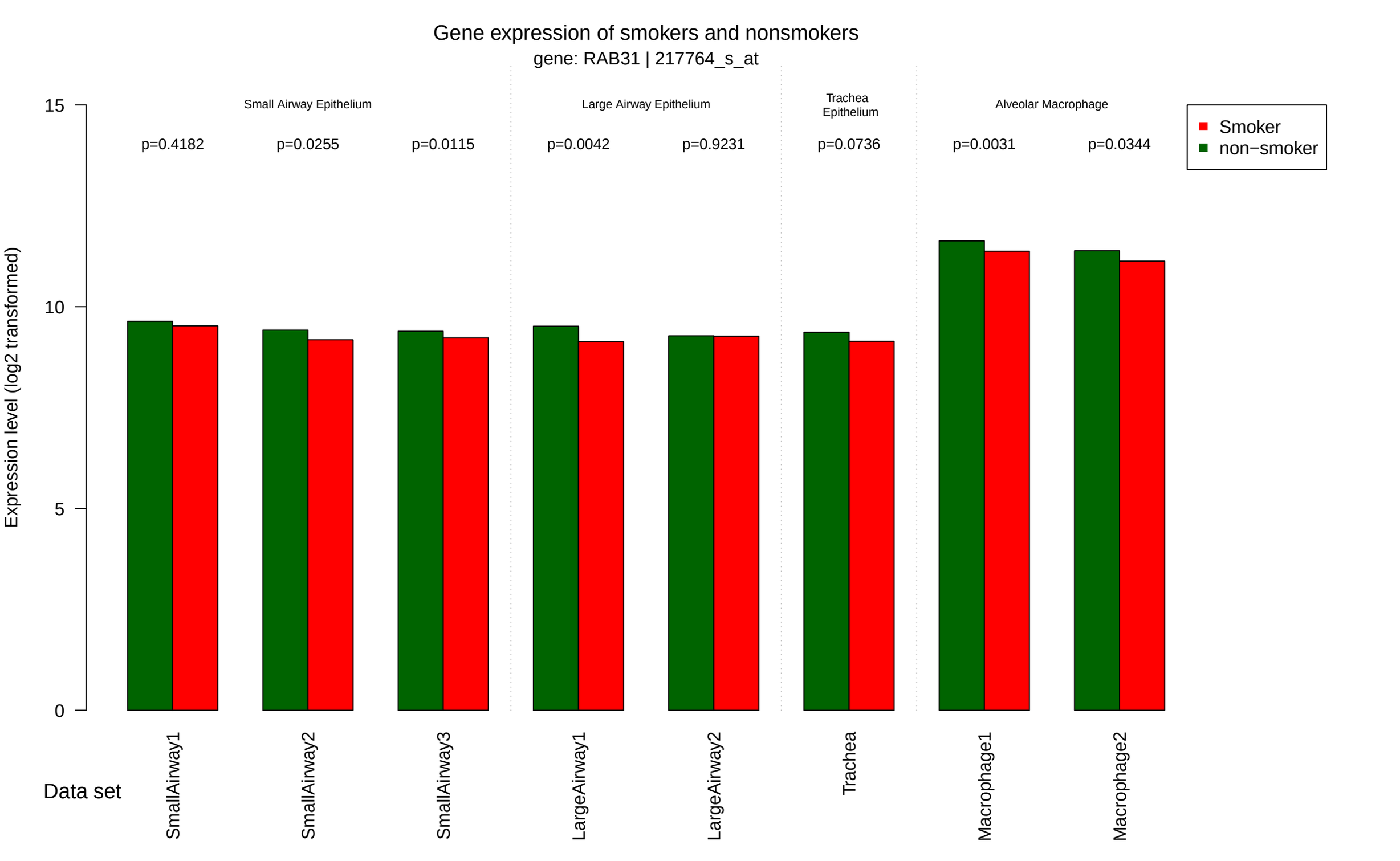
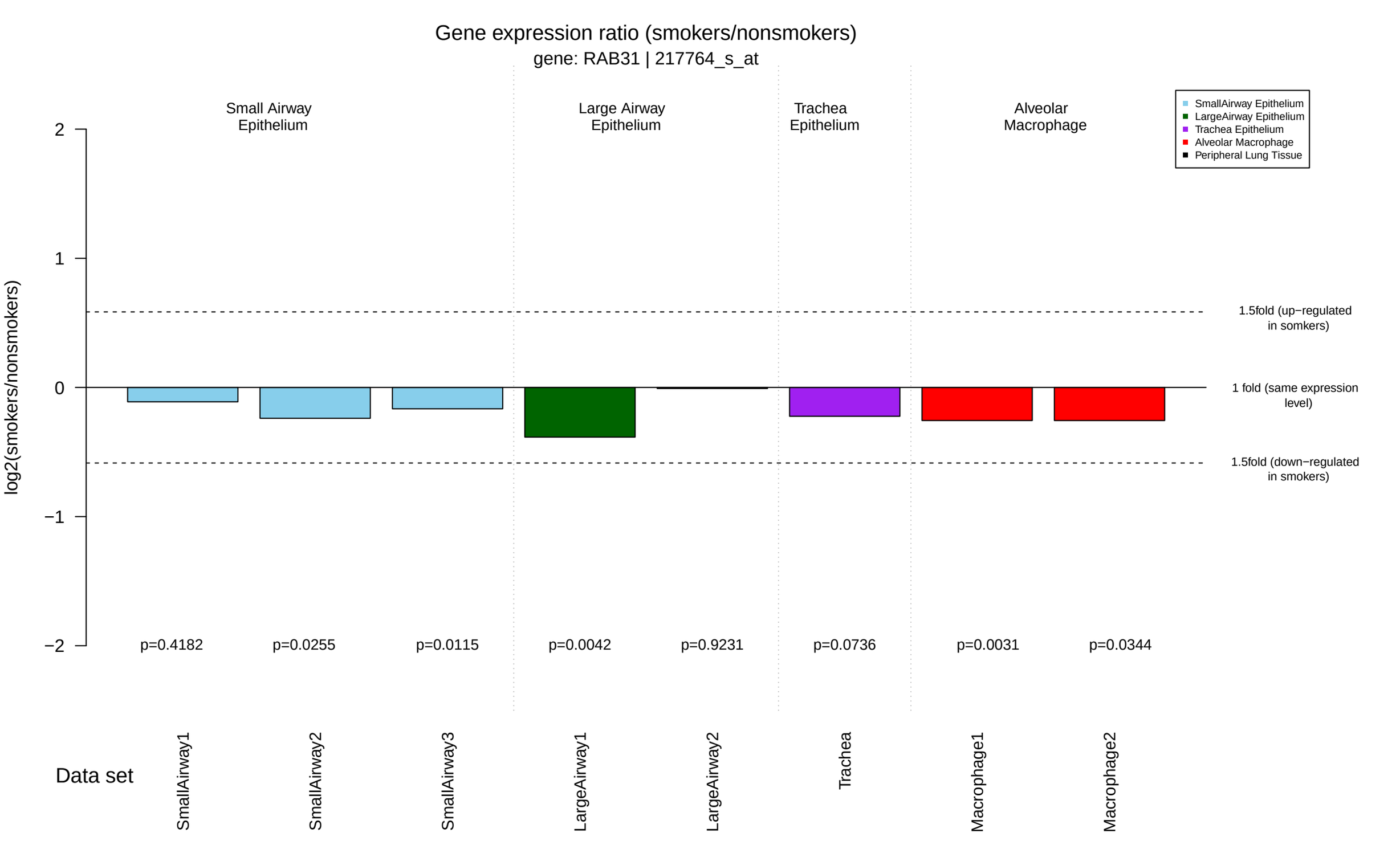
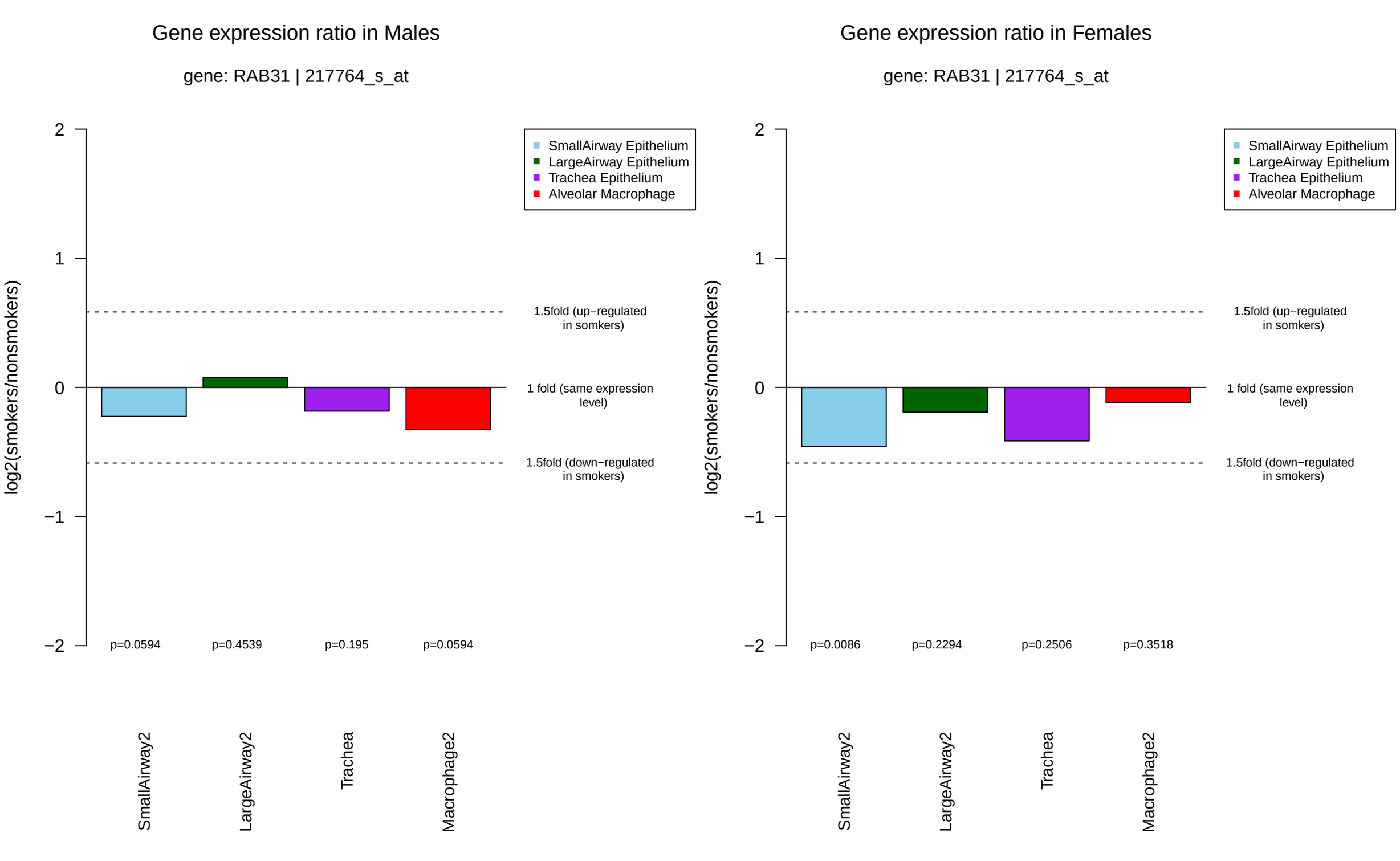
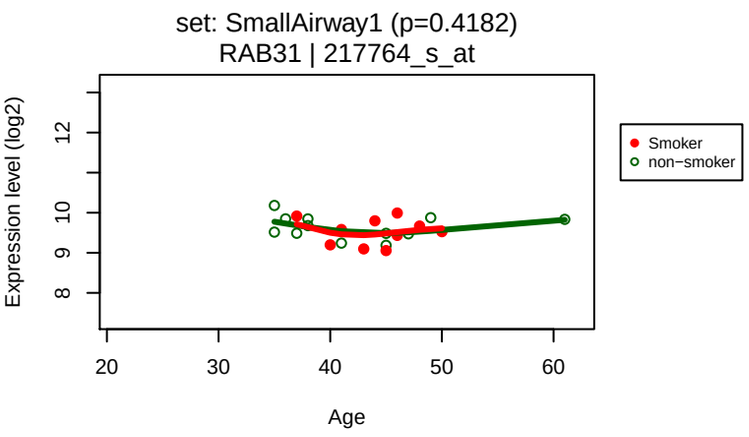
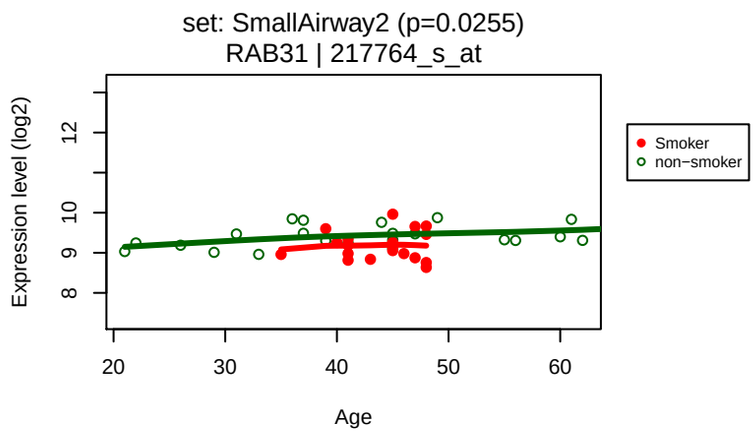
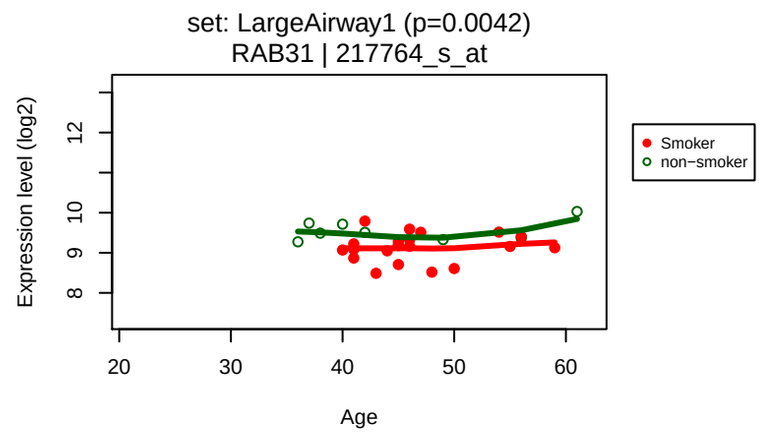

Gene expression of smokers and nonsmokers
gene: RAB31 | 217764_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4182
p=0.0255
p=0.0115
p=0.0042
p=0.9231
p=0.0736
p=0.0031
p=0.0344
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

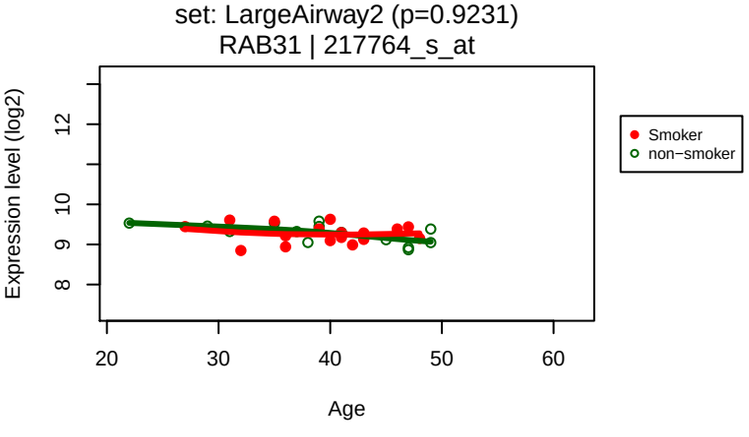

Gene expression ratio (smokers/nonsmokers)
gene: RAB31 | 217764_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4182
p=0.0255
p=0.0115
p=0.0042
p=0.9231
p=0.0736
p=0.0031
p=0.0344
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

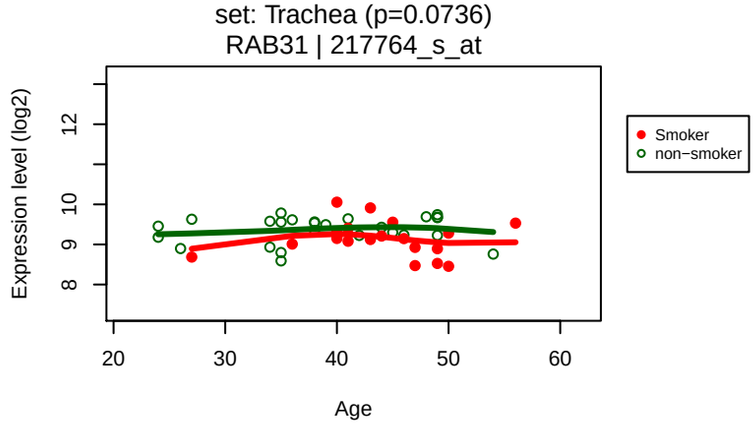

Gene expression ratio in Males
gene: RAB31 | 217764_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0594
p=0.4539
p=0.195
p=0.0594
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAB31 | 217764_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0086
p=0.2294
p=0.2506
p=0.3518
SmallAirway2
LargeAirway2
Trachea
Macrophage2

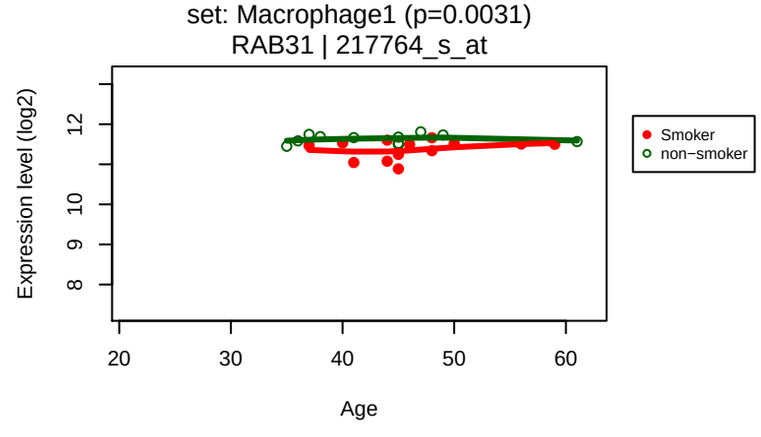

set: SmallAirway1 (p=0.4182)
RAB31 | 217764_s_at
Expression level (log2)
Age
Smoker
non−smoker

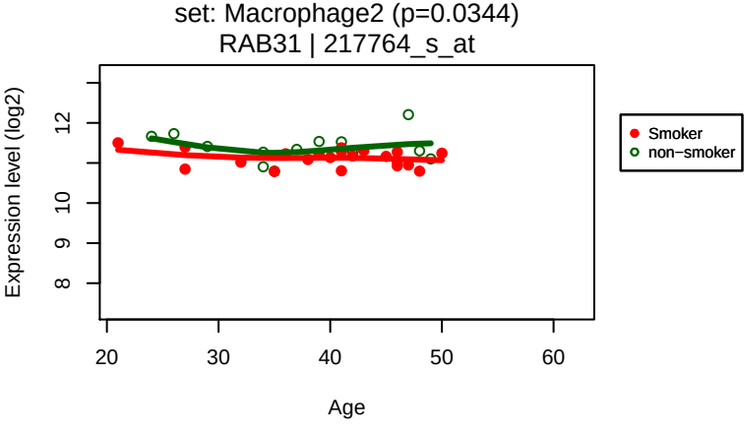

set: SmallAirway2 (p=0.0255)
RAB31 | 217764_s_at
Expression level (log2)
Age
Smoker
non−smoker

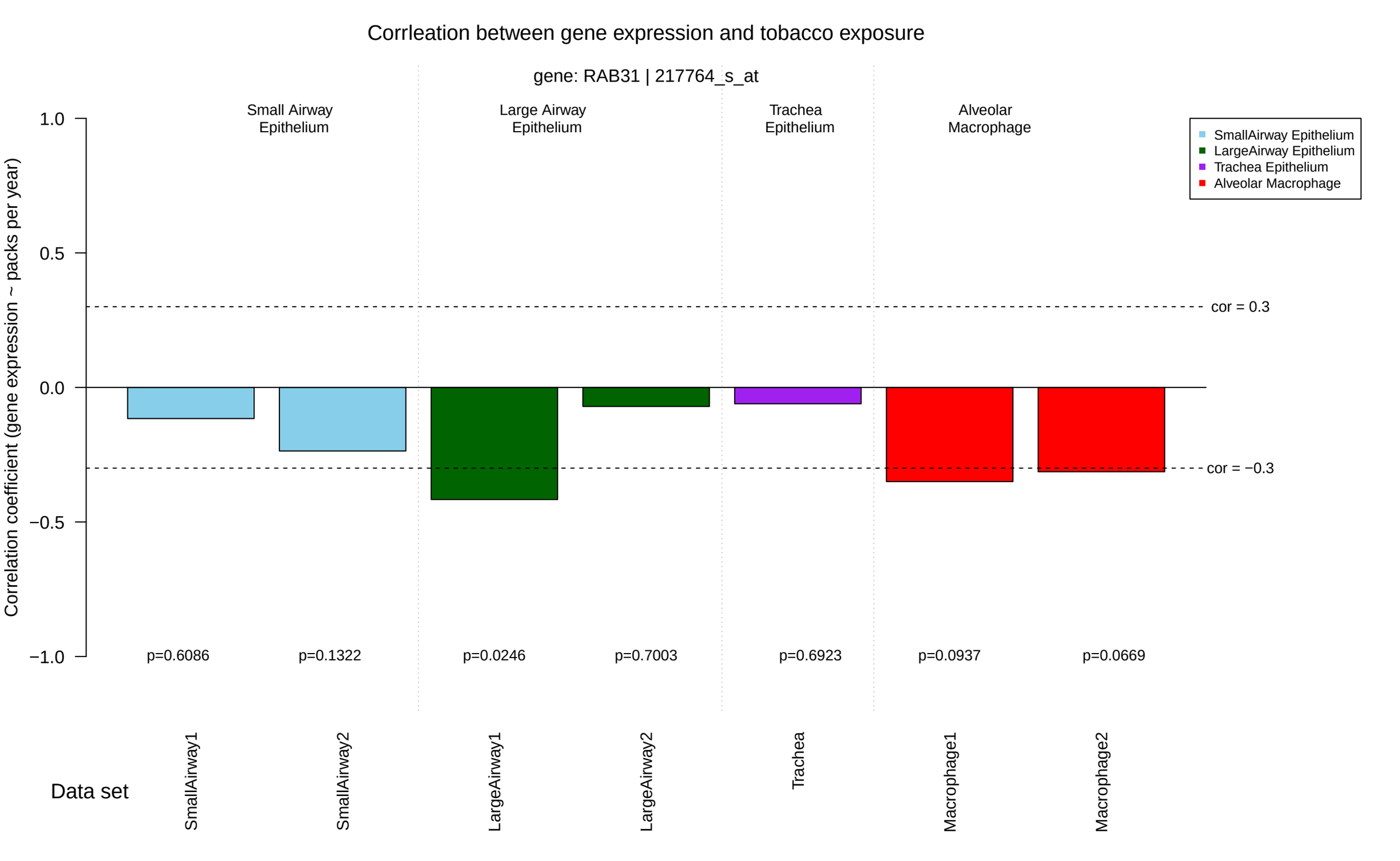

set: LargeAirway1 (p=0.0042)
RAB31 | 217764_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9231)
RAB31 | 217764_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0736)
RAB31 | 217764_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0031)
RAB31 | 217764_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0344)
RAB31 | 217764_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAB31 | 217764_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.6086
p=0.1322
p=0.0246
p=0.7003
p=0.6923
p=0.0937
p=0.0669
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2